

Optimization – 2

Pointwise optimisation, extreme points and the bang-bang principle

Homework

10 сентября 2026 года

Theoretical minimum. Extreme points and the Krein–Milman theorem; a linear functional attains its extrema at extreme points and a convex one attains its maximum there; the extreme points of the simplex, of the L^∞ ball, of $\{0 \leq f \leq 1, \int f = c\}$ and of the doubly stochastic matrices; the pointwise principle and the multiplier attached to one integral constraint; the bang-bang form of the answer.

1. Prove that the extreme points of $\{f : 0 \leq f \leq 1, \int_0^1 f = c\}$ are exactly the indicators of sets of measure c , and solve $\max \int_0^1 wf$ over this class for a continuous w .
2. (Putnam 2025) Find the largest real a and the smallest real b such that $ax(\pi - x) \leq \sin x \leq bx(\pi - x)$ for all $x \in [0, \pi]$.
3. (Proposed problems) Suppose $a, b, c, d \geq 0$ have arithmetic mean 1; then their quadratic mean q lies in $[1, 2]$. For $q \in [1, 2]$ let M_q be the largest possible geometric mean of such a, b, c, d . Determine M_q and prove that $M_q + q \geq 2$.
4. (Proposed problems) Find the best lower and upper bounds for $\sum_{i=1}^n \cos(\angle A_i)$ over all convex n -gons $A_1 A_2 \dots A_n$.
5. (SEEMOUS 2010) Inside a square consider circles such that the sum of their circumferences is twice the perimeter of the square.
 - a) Find the minimum number of circles with this property.

- b) Prove that there are infinitely many lines meeting at least three of these circles.
6. Prove that a convex function on a polytope attains its maximum at a vertex, and deduce that $\max \sum_{i < j} |x_i - x_j|$ over the cube $[0, 1]^n$ is attained at a vertex; find its value.
 7. Prove the pointwise principle of Fact 1 carefully, including the measurability of the maximiser, and give an example where a single constraint $\int f = 0$ makes the naive pointwise answer inadmissible and a multiplier appears.
 8. Describe the extreme points of the unit ball of $C[0, 1]$ and prove that the unit ball of $L^1[0, 1]$ has no extreme points at all. Explain why this does not contradict the Krein–Milman theorem, and what it means for the attainability of a supremum over such a ball.
 9. Prove that the maximum of a linear functional over the probability measures on a compact set is attained at a Dirac measure, and use this to find $\max \int x^2 d\mu$ over probability measures on $[-1, 1]$ with $\int x d\mu = 0$.
 10. Prove that $\max \int_0^1 fg$ over all f with $|f| \leq 1$ and $\int_0^1 f = 0$ equals $\min_{c \in \mathbb{R}} \int_0^1 |g(x) - c| dx$, and identify the optimal f and the optimal c for $g(x) = x^2$.